GUL HAMEED

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SUMMARY

Modelling Specialist at AstraZeneca, developing numerical tools for process modelling, with a PhD (thesis submitted) on trust-region algorithms for grey-box optimisation from the University of Surrey. Before AstraZeneca I built DECO2 and B-HYPSYS, open-source optimisation tools used with government and industry partners: mixed-integer and nonlinear problems with 50,000+ constraints, and hourly dispatch with ~150,000 variables across 8,760 periods. I contributed the funnel globalisation mechanism to Pyomo's trust-region solver (released in Pyomo 6.10.1) and have nine first-author papers. My strength is making large, badly behaved models solve reliably and repeatably for the people who use them.

TECHNICAL SKILLS

- **Optimisation and OR:** LP, MILP, MINLP, NLP; scheduling and dispatch over long horizons; derivative-free, surrogate-based and trust-region methods; benchmarking with performance profiles
- **Solving large instances:** custom initialisation and warm starts, structured multistart, homotopy continuation, model reduction and surrogate substitution; diagnosing infeasible or badly scaled models; sensitivity analysis
- **Parameter estimation and process modelling:** estimability analysis, design space identification, first-principles and grey-box models, process flowsheets with embedded PDAE models
- **Solvers:** Gurobi, CPLEX, SCIP, BARON, IPOPT (MA27), DICOPT, CONOPT; benchmarked my PhD solvers against NOMAD, Nelder-Mead, COBYQA and NEWUOA
- **Programming:** Python (primary, production quality), Pyomo, MATLAB, GAMS; foundational C++
- **ML and statistics:** Gaussian process regression, supervised learning, statistical modelling, surrogate models for expensive simulations
- **Software engineering:** professional Git/GitHub workflow (version control, branching, pull requests, code review) at AstraZeneca and throughout the PhD; modular framework design, pytest, technical documentation, HPC; Pandas, NumPy, Plotly, Matplotlib, Streamlit

EXPERIENCE

AstraZeneca, Macclesfield, UK **Sep 2026 - Present**
Modelling Specialist

- Developing model-agnostic numerical tools for process modelling: sensitivity analysis, parameter estimation and design of experiments

University of Surrey, Guildford, UK **Mar 2022 - Aug 2026**
Scientific Software Developer, Frameworks and Simulators

- Built DECO2, open-source MINLP/MILP decision-support tools for energy-system planning (Pyomo; Gurobi, SCIP, BARON, DICOPT) scaling to 5,000+ binary and 20,000+ continuous variables with 50,000+ constraints through custom two-stage initialisation; deployed in case studies for Malaysia, the Philippines, Mauritius and Pakistan
- Wrote the proposals for, and as lead researcher delivered, the £70k RIDA project (FCDO and Research England ODA) behind DECO2, including keynotes and workshops on energy decarbonisation that I delivered and organised for government and industry in Kuala Lumpur, Manila and Guildford (2024)
- Built B-HYPSYS, a dynamic dispatch and design framework scheduling hourly operation over a full year (8,760 periods, ~150,000 variables and constraints) with storage, ramp-rate and turndown limits; made the nonconvex NLP solve reliably with staged homotopy and a 21-point structured multistart, and shipped it to non-developers behind an Excel interface
- Building a process simulator from scratch in Python (Streamlit, Pandas, Plotly): modular architecture, custom NLP and trust-region solver orchestration, pytest coverage throughout

Carnegie Mellon University, Pittsburgh, USA **Jan - Mar 2026**
Visiting Researcher, Prof Lorenz T Biegler's group

- Contributed the funnel globalisation mechanism to Pyomo's trust-region solver (PR #3832, released in Pyomo 6.10.1); the first solver contribution to Pyomo from the University of Surrey
- Applied trust-region grey-box optimisation to carbon capture and utilisation flowsheets with embedded PDAE models (dual-function material reactors, pressure swing adsorption)

University of Surrey - Fluor Pilot Plant, Guildford, UK **Sep - Dec 2025**
Plant Manager

- Trained undergraduate and master's students to operate a pilot-scale process plant and supervised their experimental campaigns; owned safety, scheduling and data collection

EARLIER EXPERIENCE

- Research Assistant, Tandem Catalysis (RSC-funded), University of Surrey, 2022-2023: first-author paper on Ni-phosphide RWGS catalysts (Journal of CO2 Utilisation)
- Graduate Teaching Assistant, University of Surrey, 2023-2024: heat transfer laboratory and advanced pharmaceutical formulations
- Graduate Trainee Officer, Oil & Gas Development Company (OGDCL), Pakistan, 2021; Trainee Engineer, WellServe Oilfield Services, Pakistan, 2020

EDUCATION

University of Surrey **Jul 2023 - Oct 2026**
PhD, Chemical & Process Systems Engineering (thesis submitted)

- Trust-region (filter and funnel) algorithms for surrogate-based **nonlinear grey-box optimisation** combining first-principles models with data-driven surrogates, with convergence proofs and process and energy case studies. Supervisor: Prof Michael Short
- Benchmarked against classical derivative-free solvers using performance profiles, with pytest coverage and HPC runs

University of Surrey **Feb 2022 - Feb 2023**
MSc, Renewable Energy Systems Engineering (Distinction, 1st in class)

- Project: large-scale regional energy-planning MINLP models solved with SCIP and DICOPT

Pakistan Institute of Engineering & Applied Sciences (PIEAS) **Sep 2016 - Jul 2020**
BS, Mechanical Engineering (CGPA 3.26 / 4.00)

SELECTED PUBLICATIONS (9 FIRST-AUTHOR)

- "A trust-region funnel algorithm for grey-box optimisation." *AIChE Journal* e70258 (2026)
- "Trust-region filter algorithms utilising Hessian information for grey-box optimisation." *AIChE Journal* e70236 (2026)
- "Optimisation of synthetic natural gas production via direct air capture and utilisation using reduced models under a novel trust-region funnel method." *Systems and Control Transactions* 5 (2026)
- "Pathways to decarbonise the power sector in Mauritius." *Energy Strategy Reviews* 62 (2025): 101930
- "Optimal transition pathways to a net-zero power grid in the Philippines." *Energy Strategy Reviews* 62 (2025): 101893
- "A novel mathematical model to incorporate carbon trading and emission reduction techniques within energy planning models." *Sustainable Production and Consumption* 40 (2023): 571-589

AWARDS & ACTIVITIES

- Vice-Chancellor's Award, Postgraduate Researcher of the Year 2024: Faculty of Engineering and Physical Sciences winner and University finalist
- Finalist, IChemE Young Engineers Awards for Innovation and Sustainability (Energy and Climate Action), 2026
- Talks: SIAM Conference on Optimization (OP26), Edinburgh, 2026; ESCAPE36, Sheffield, 2026. Posters: ESCAPE34-PSE24, Florence, 2024; CAPD Annual Meeting, Pittsburgh, 2026
- Founder and first President, Student Energy Surrey, 2024-2025; Chair of the Organising Committee, Surrey PGR/ECR Conference 2025 (100+ talks across 21 schools); session chair at many university conferences
- President, PIEAS Literary Society (2019-2020), overseeing the university's non-technical societies: dramatics, arts, literature, videography and photography